NAME : DARSHINI DINESHKUMAR ROLL NO : 250701124

EXPERIMENT NO 1 .A

1.a. Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib for visualization.

AIM:

To analyze the trend of data science job postings over the last decade.

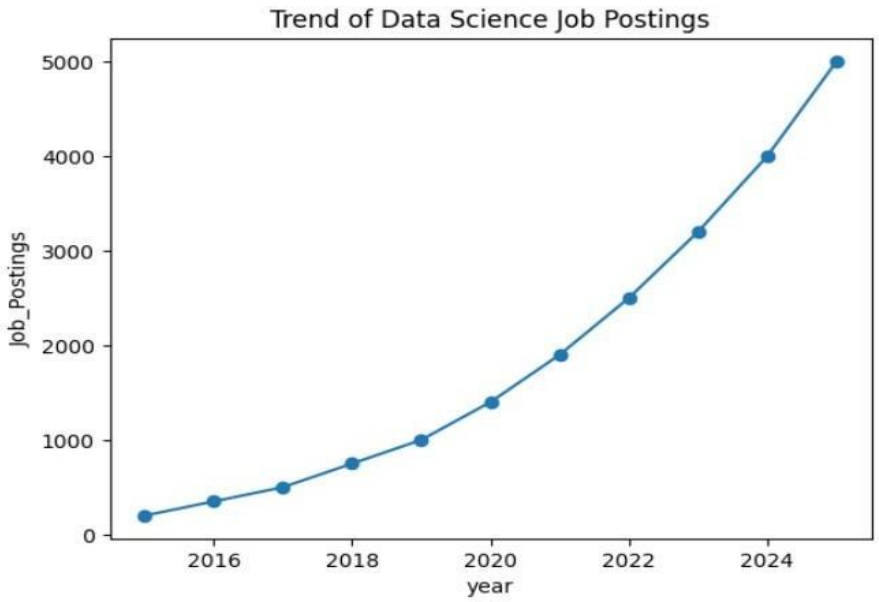
DESCRIPTION:

Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib/seaborn for visualization.

PROGRAM:

```
import pandas as pd import
matplotlib.pyplot as plt data={
   'year':list(range(2015,2026)),'Job_Postings':[200,350,500,750,1000,1400,1900,2500,3200,4000,5000]
}
df=pd.DataFrame(data)
plt.plot(df["year"],df["Job_Postings"],marker='o')
plt.xlabel("year") plt.ylabel("Job_Postings")
plt.title("Trend of Data Science Job Postings") plt.show()
```

OUTPUT:



RESULT:

Job postings increased every year become much faster after 2020 overall trend is upward showing high demand data Science jobs.

EXPERIMENT NO 1 .B

1.b Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

AIM:

To Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

DESCRIPTION:

Use a dataset of job postings and categorize them into different roles. Visualize the distribution using pie charts or bar plots.

PROGRAM:

```
import matplotlib.pyplot as plt

Roles=['Data Analyst','Data Engineer','Data Scientist','ML Engineer','Business Analyst']

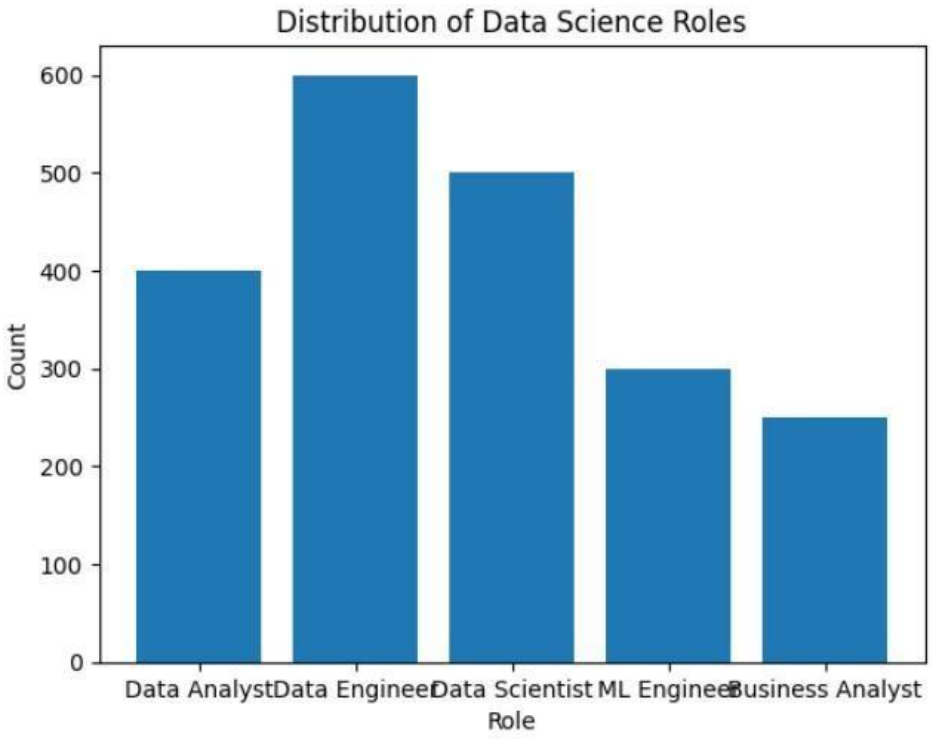
counts=[400,600,500,300,250] plt.bar(Roles,counts)

plt.title("Distribution of Data Science Roles")

plt.xlabel('Role') plt.ylabel('Count')

plt.show()
```

OUTPUT:



RESULT:

Data science is occusing role in the dataset (count=3) Data Analyst (3),Data engineer (2) and ML engineer are equally dataset (count=1).

EXPERIMENT NO 1 .C

1.c.Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

AIM:

Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

DESCRIPTION:

Create small datasets for each type and explain their characteristics.

PROGRAM:

```
import pandas as pd structured_data=pd.DataFrame({
   'ID':[101,102,103],
   'Name':['David','Emma','John'],
   'Age':[28,32,27]
})
print("Structured Data:\n",structured_data)
unstructured_data="This is a Sample document containing text,images and vedio content."
print("\n Unstructed Data:\n",unstructured_data) semi_structured_data={
   'ID':101,
   'Name':'David',
   'Attributes':{'Height':170,'Weight':65}
}
print("\n Semi structured data:\n",semi_structured_data)
```

OUTPUT:

```
Structured Data:
     ID   Name  Age
0  101  David   28
1  102   Emma   32
2  103   John   27

 Unstructed Data:
 This is a Sample document containing text,images and vedio content.

 Semi structured data:
 {'ID': 101, 'Name': 'David', 'Attributes': {'Height': 170, 'Weight': 65}}
```

RESULT:

Structured data (tabular, fixed schema) is easiest to query via SQL. Semi-structured (JSON/XML) is self-describing but schema-flexible. Unstructured data (free text) carries no explicit schema.

EXPERIMENT NO 1 .D

1.D Conduct an experiment to encrypt and decrypt given sensitive data.

AIM:

Conduct an experiment to encrypt and decrypt given sensitive data.

DESCRIPTION:

Use the cryptography library to encrypt and decrypt a piece of data.

PROGRAM:

```
from cryptography.fernet import Fernet
key=Fernet.generate_key() cipher_suite=Fernet(key)
plain_text=b"Computer Science Engineering"
cipher_text=cipher_suite.encrypt(plain_text)
decrypted_text=cipher_suite.decrypt(cipher_text)
print("Original data :",plain_text) mid =
len(cipher_text) // 2
print("Encrypted data :", cipher_text[:mid]) print("
", cipher_text[mid:]) print("Decrypted data
:",decrypted_text)
```

OUTPUT:

```
Original data : b'Computer Science Engineering'
Encrypted data : b'gAAAAABqgEaJaJgGS36ygzNQp_WQBpeS_RmZFdRBqZgjIdCj3jO1lZvuxPjJ'
                 b'tzc1j3kBJABDw8SZ80RZWkn33utNYEJ7ADT8J5P4-eFECo4EFoFiCti2FgY='
Decrypted data : b'Computer Science Engineering'
```

RESULT:

The Fernet symmetric-key algorithm successfully encrypted and decrypted the original data text, verifying that any tampering with the ciphertext would cause decryption to fail.